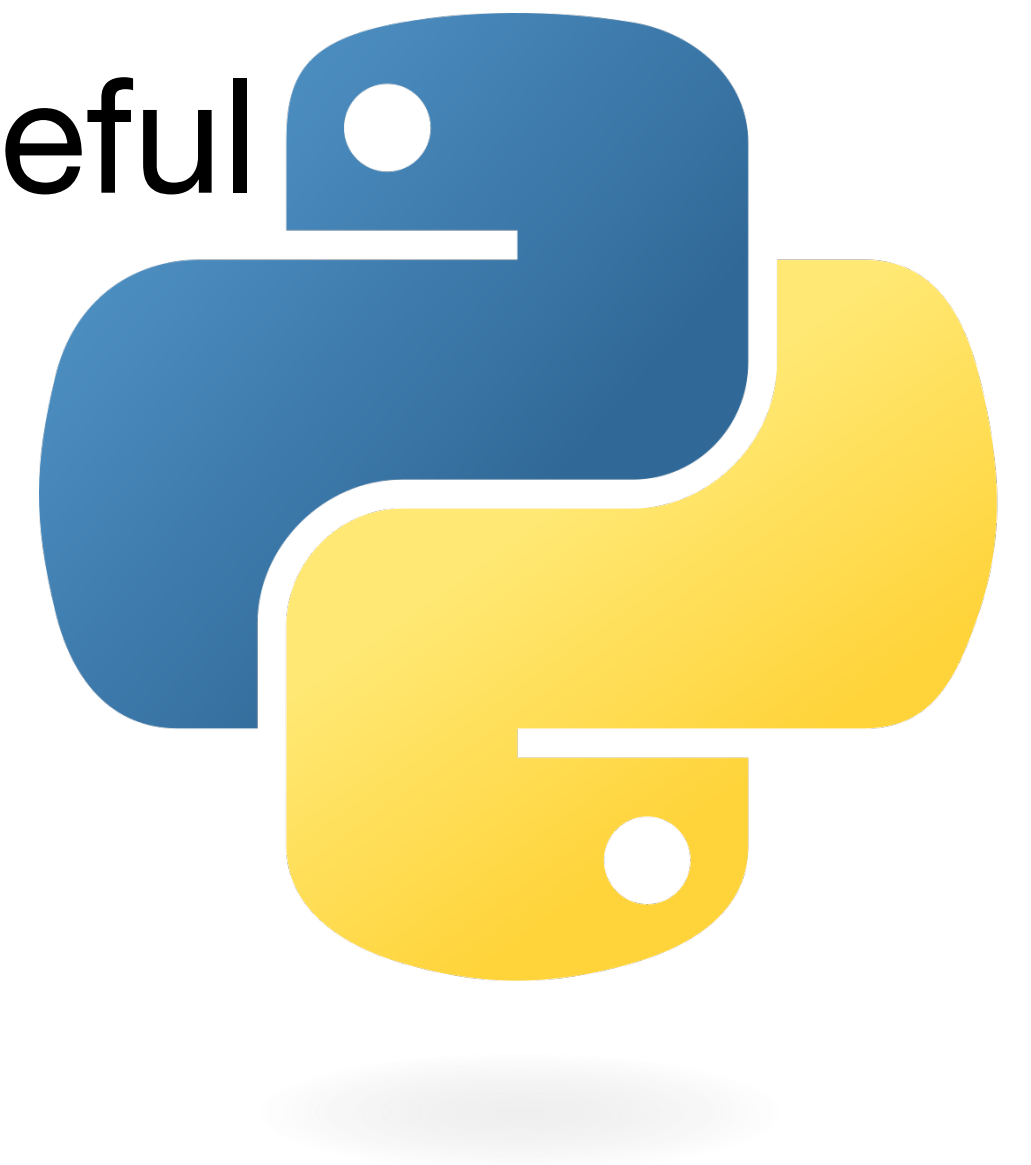


Motivating question

- Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, how can we determine values $x \in \mathbb{R}$ (if they exist), such that $f(x) = 0$?
- What assumptions do we need to make of f ?
- If f was a polynomial, then factoring f gives us useful information about the roots of f .
- However, not all polynomials factor easily,
and not all interesting functions are polynomials.

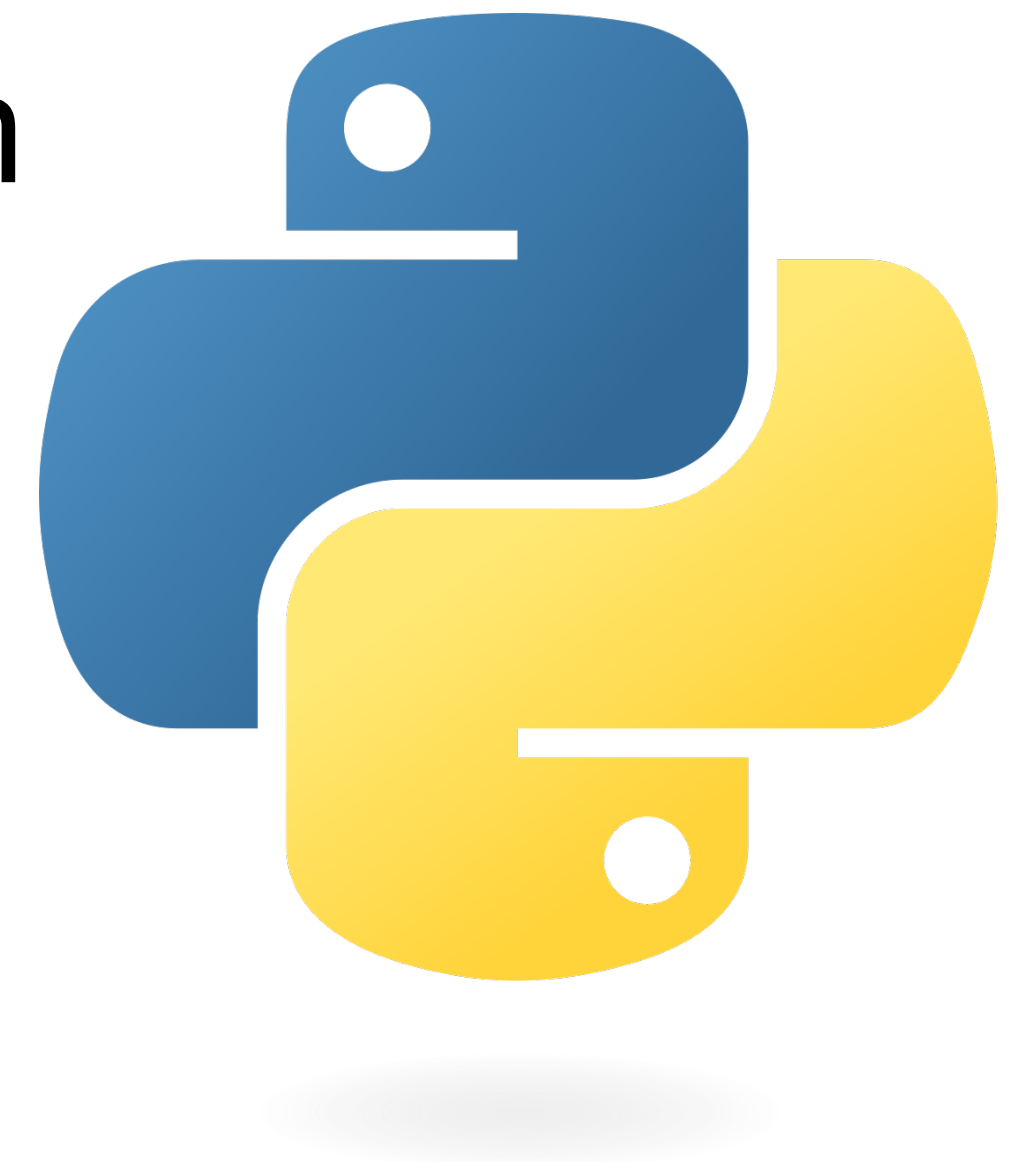


Motivation continued

- For instance, consider the function g as defined below,

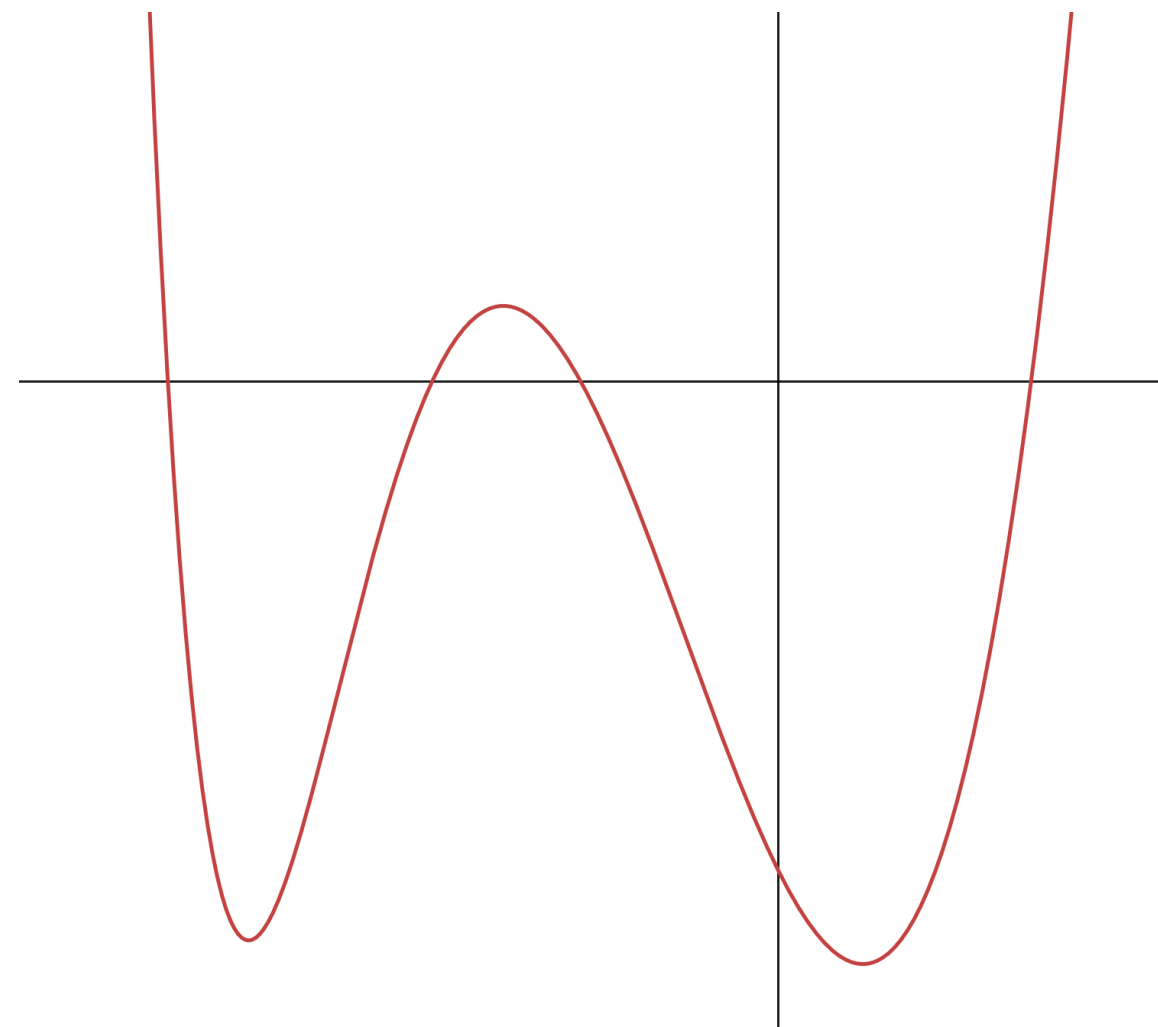
$$g(x) = \begin{cases} 0 & \text{if } x = \pi \\ 2 & \text{else} \end{cases}$$

- It may be very hard to find the roots of a function like g using Python (in fact g has *no roots* in Python - think about this).



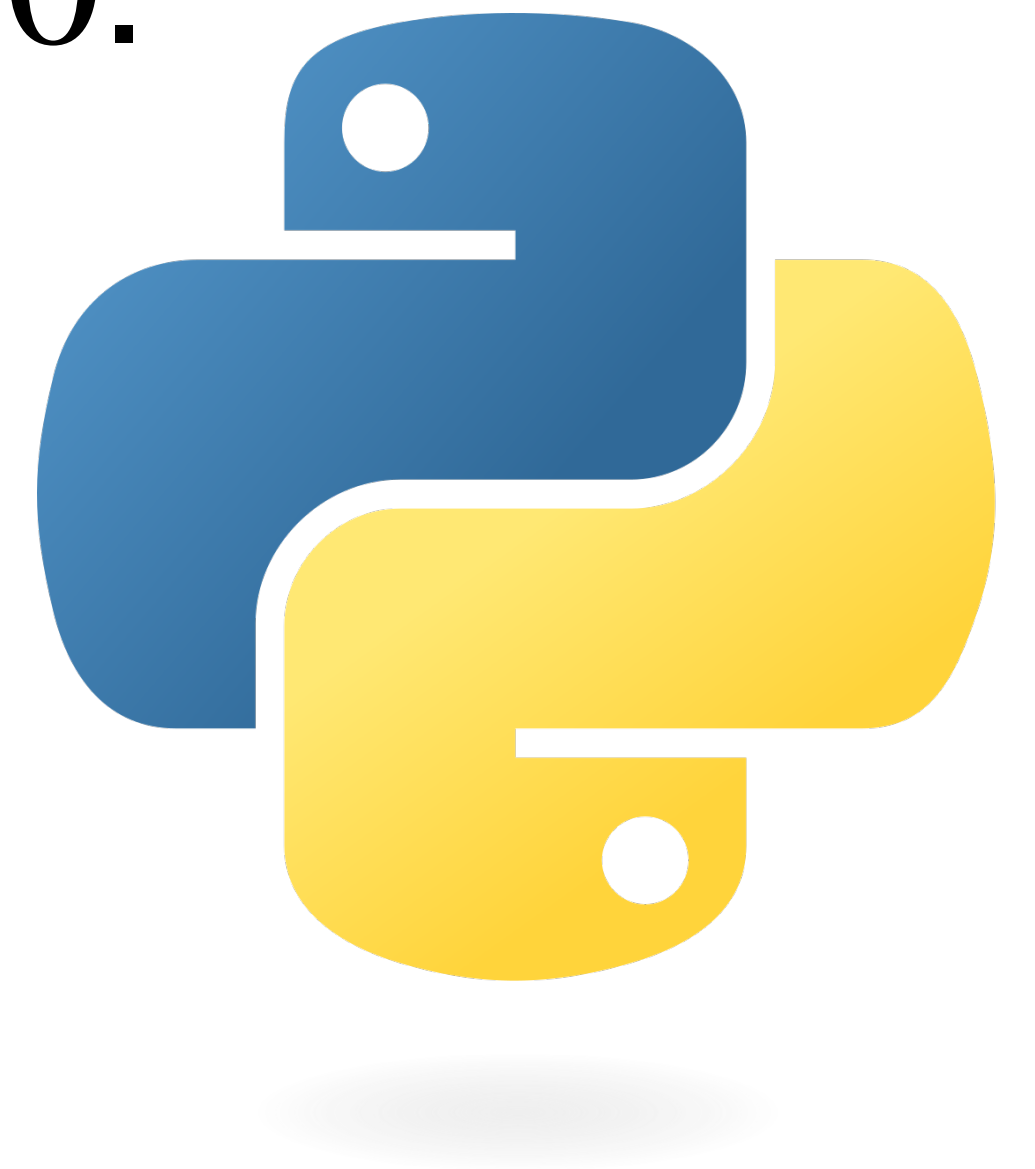
An important tool

- Throwing in the hypothesis that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous gives us a key tool from calculus I. What is it called?
- **Hint:** One can think of a continuous function as a function that can be drawn without lifting the pencil off of the page.



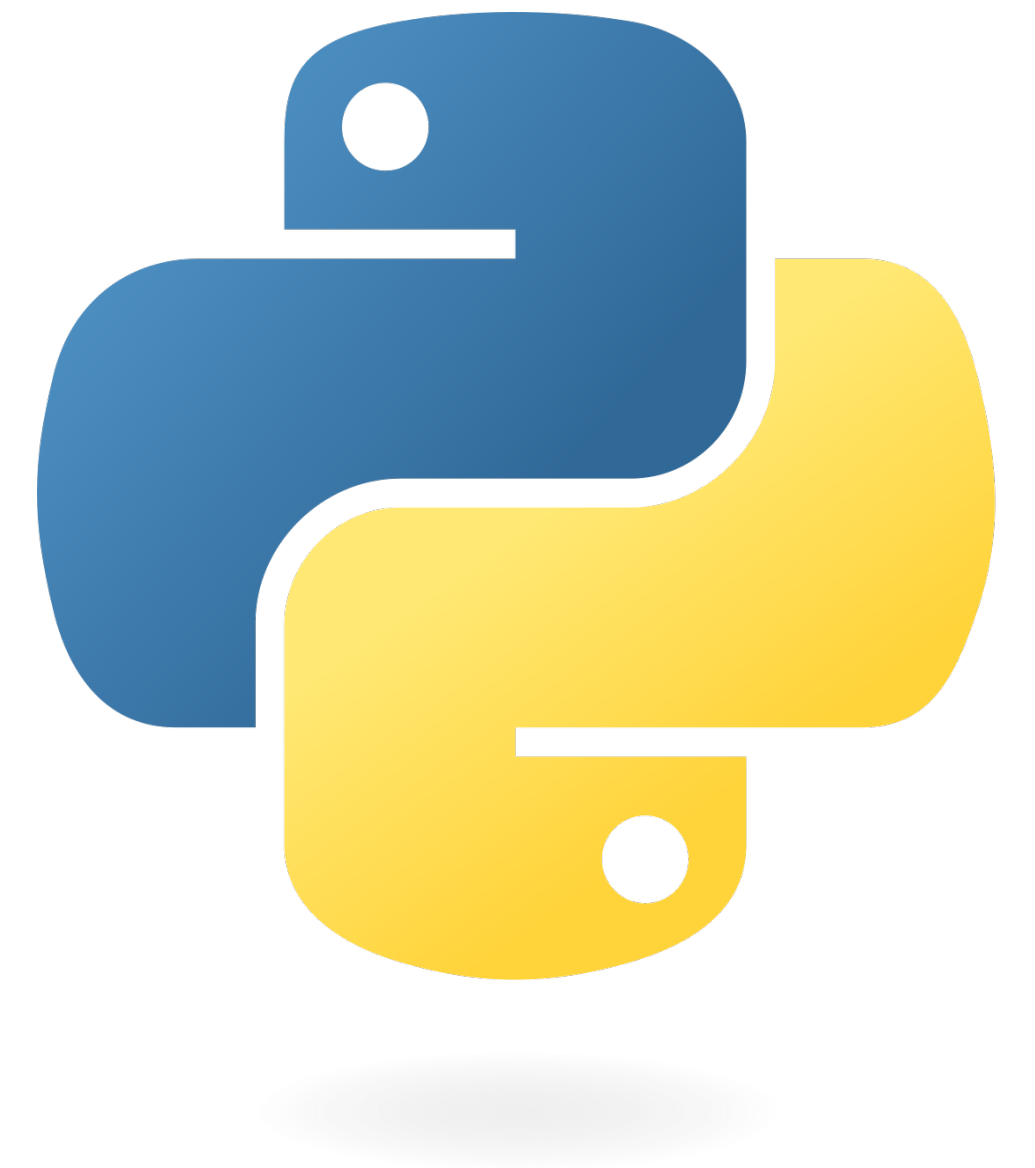
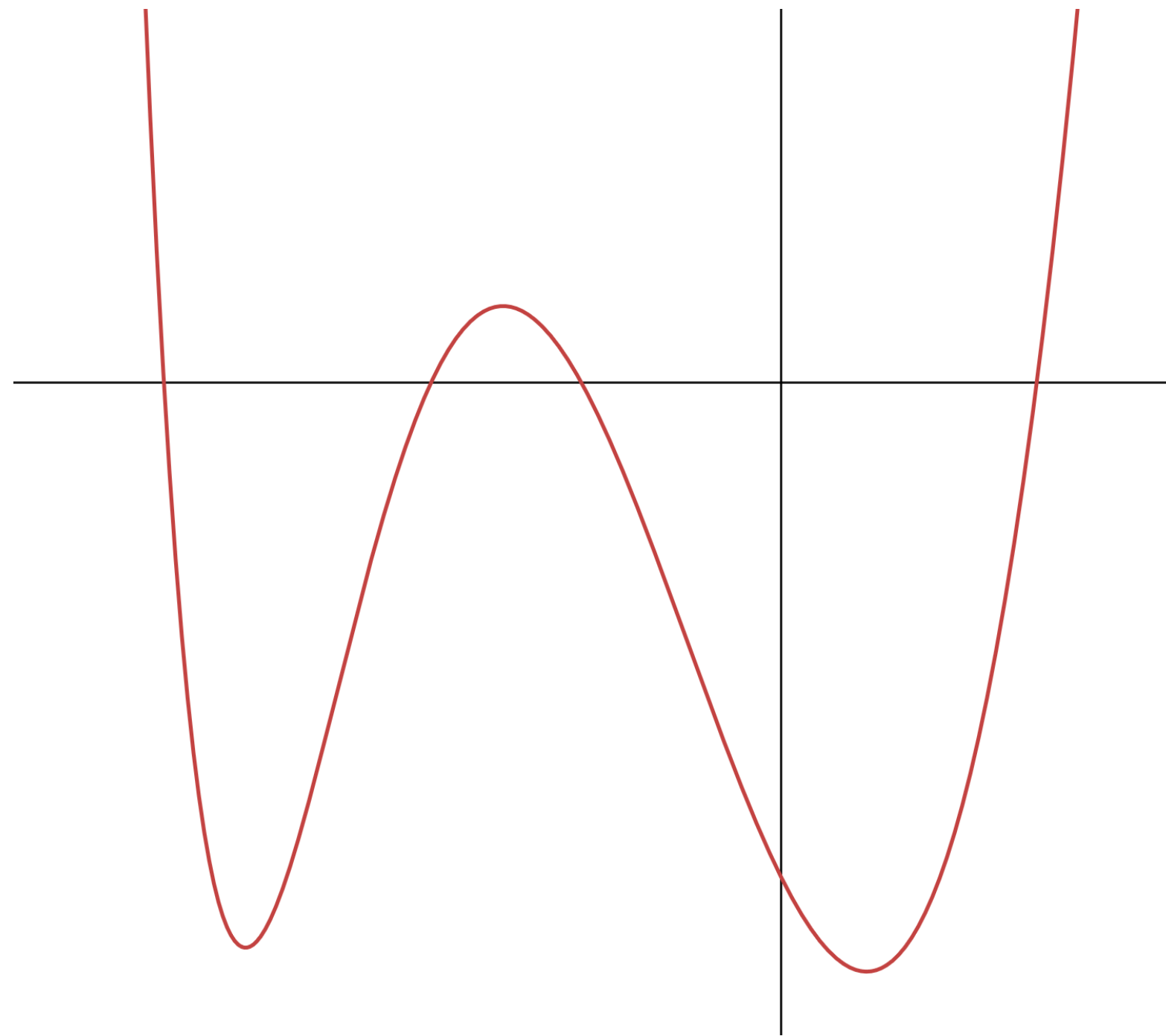
Intermediate value theorem (IVT)

- Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and let $[a, b] \subseteq \mathbb{R}$. If there exists $c, d \in [a, b]$, such that $f(c) \leq \alpha \leq f(d)$, then there exists $x_0 \in [c, d]$, such that $f(x_0) = \alpha$.
- This theorem is most interesting to us when $\alpha = 0$.



Constructing an algorithm

Question: We have identified IVT as a useful tool, but how can it be leveraged to create an algorithm?



Bisection method - 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and let $[a, b] \subseteq \mathbb{R}$. Suppose that you previously knew (perhaps through the means of graphing f) that f changes sign on $[a, b]$ exactly once.

Let $c = (a + b)/2$ be the midpoint of $[a, b]$.

If $f(a) = 0$, $f(b) = 0$, or $f(c) = 0$, then we are done as we have already found a root of f on $[a, b]$.

Otherwise, we will refine our interval in the following way...

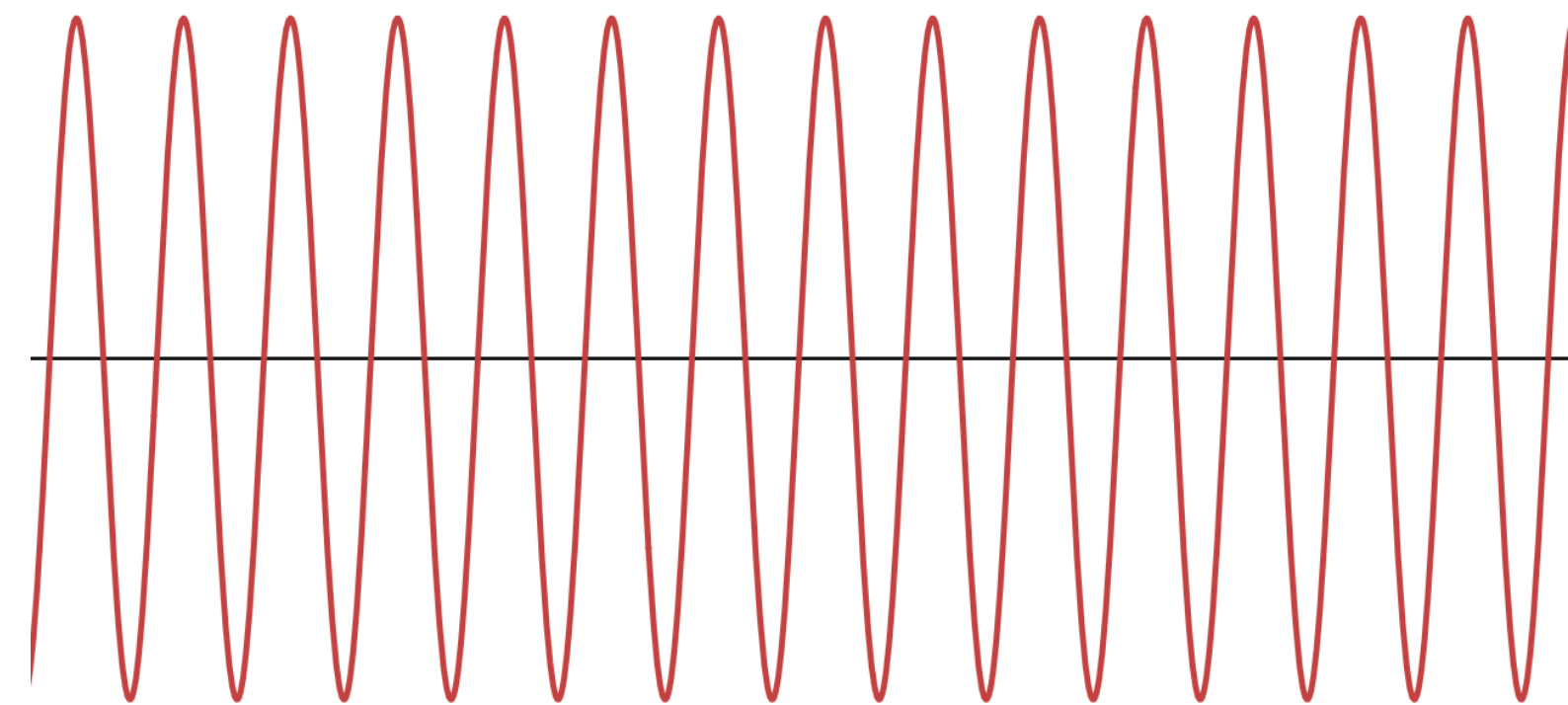
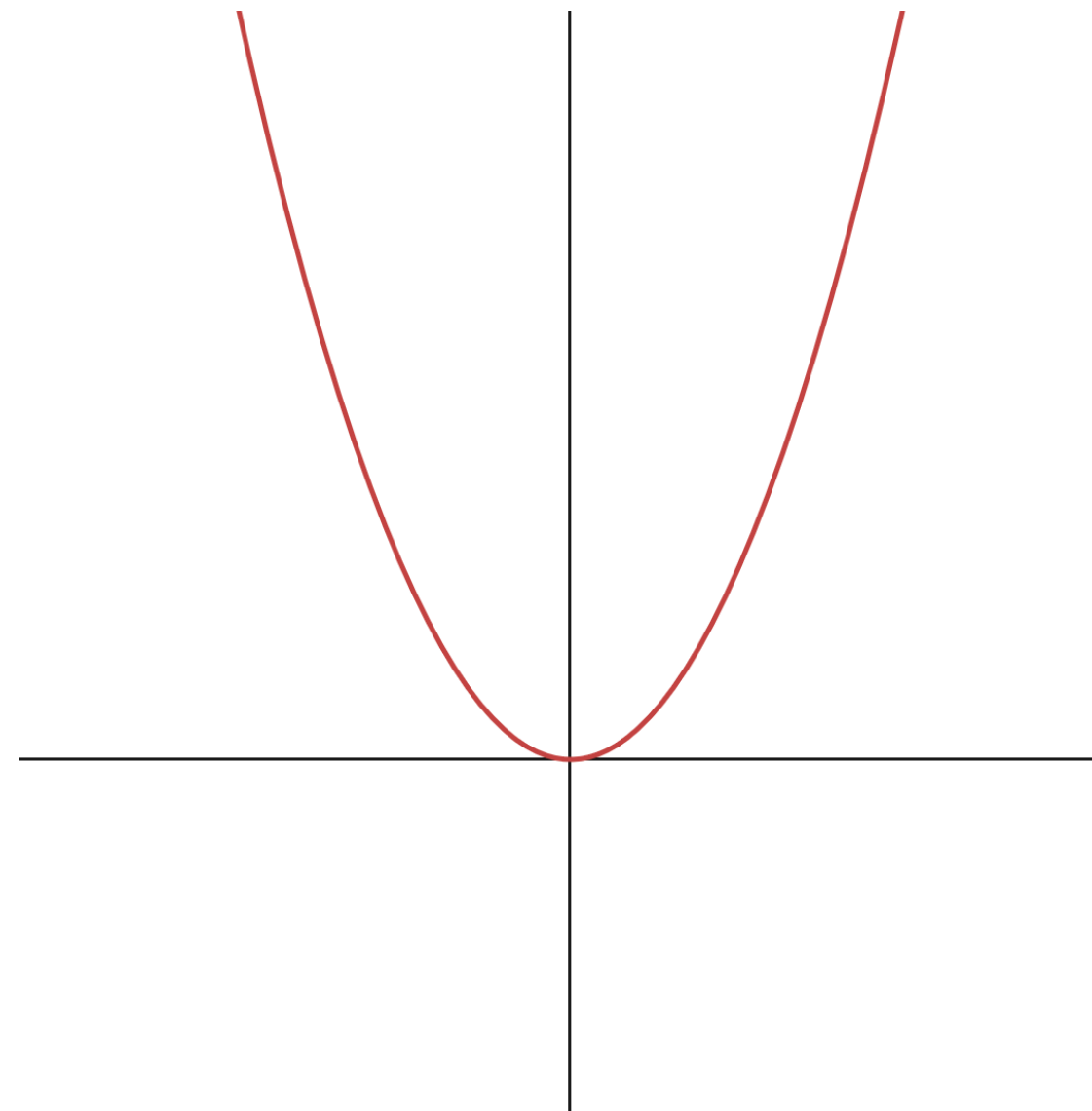
Bisection method - 2

As f changes sign exactly once on $[a, b]$ and as $f(a) \neq 0$, $f(b) \neq 0$, and $f(c) \neq 0$, then f must change sign on $[a, c]$ or $[c, b]$. If f changes sign on $[a, c]$, then $f(a) \cdot f(c) < 0$ and similarly for $[c, b]$.

If $f(a) \cdot f(c) < 0$ update $a = a$ and $b = c$. Otherwise update $a = c$ and $b = b$. Repeat the process until... (we need to choose a stopping condition)

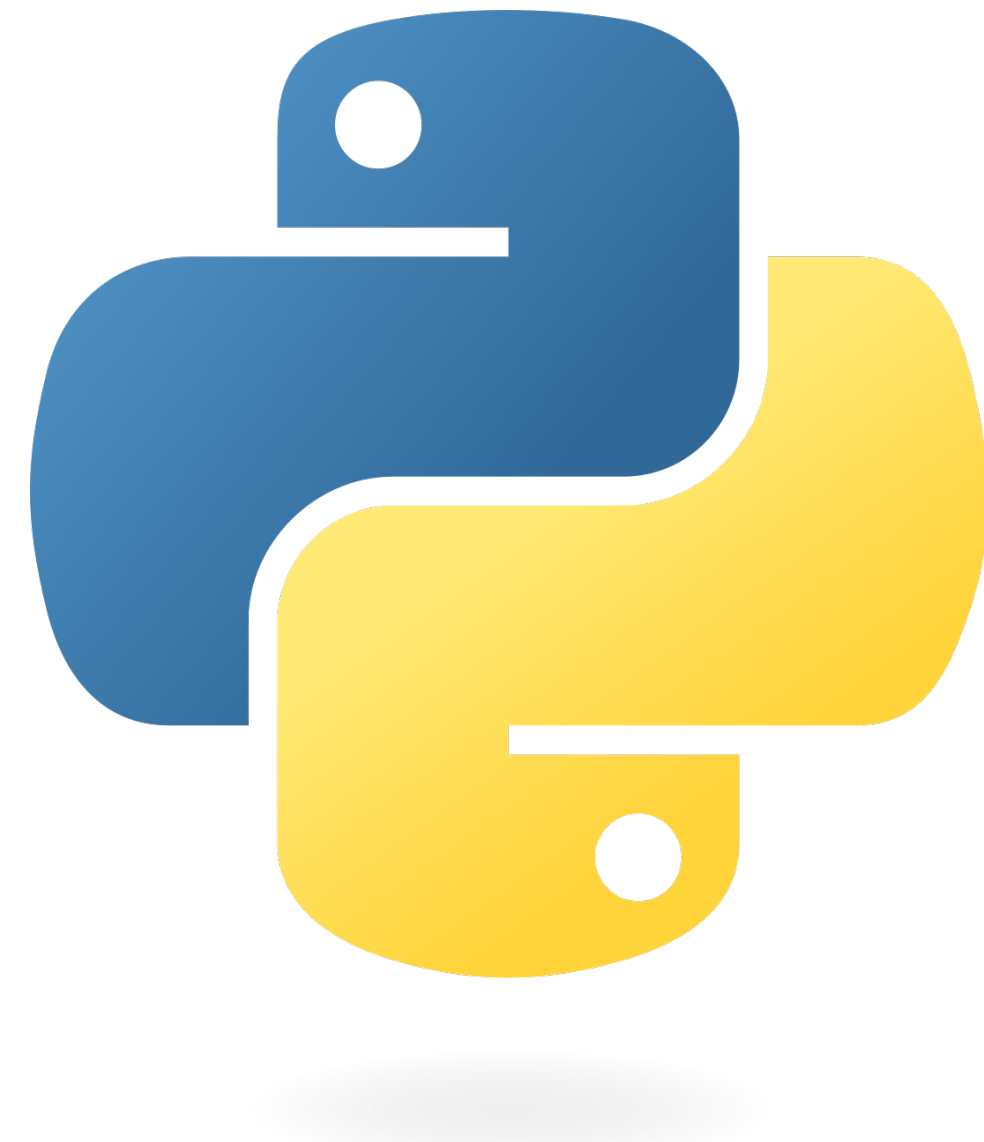
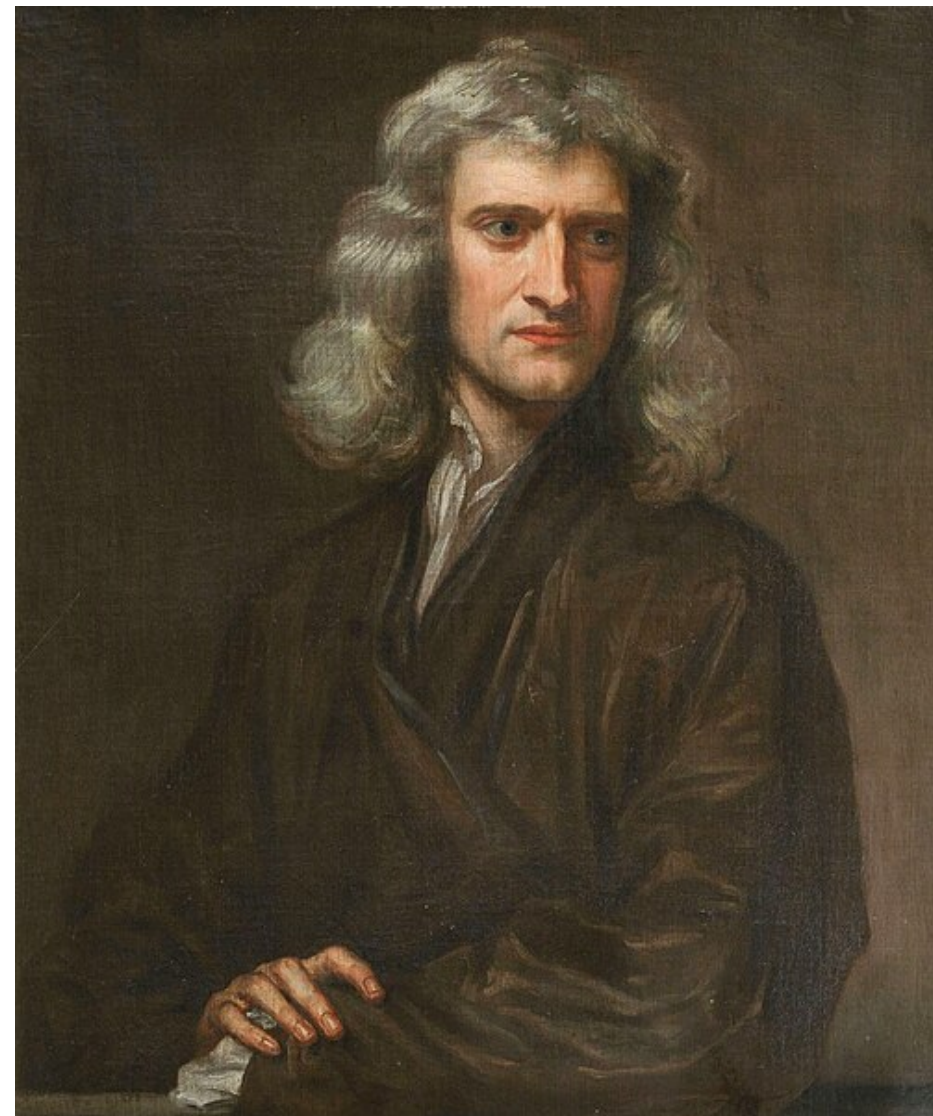
Discussion

Question: Bisection method is a simple and relatively robust root finding algorithm, but it has some key drawbacks. Identify them. (**Hint:** Consider the two images below.)



Introducing Newton's Method

- Another very popular root finding method is **Newton's method**. It gains the strength of faster convergence and the ability to detect even multiplicity roots, but has its own drawbacks (we will discuss these as they come up).



Newton's method assumptions

In order to apply bisection method we needed the following assumptions: $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and changes sign exactly once on $[a, b]$.

In order to apply Newton's method we need the following assumptions: $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $[a, b]$, f' exists everywhere on $[a, b]$, and f has a root α somewhere inside of $[a, b]$.

Note: Speed of convergence will depend heavily on whether or not $f'(\alpha) = 0$.

Newton's method - 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function with derivative on $[a, b]$ and one root $\alpha \in [a, b]$. Begin with an initial guess x_0 for where the root occurs. Let

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

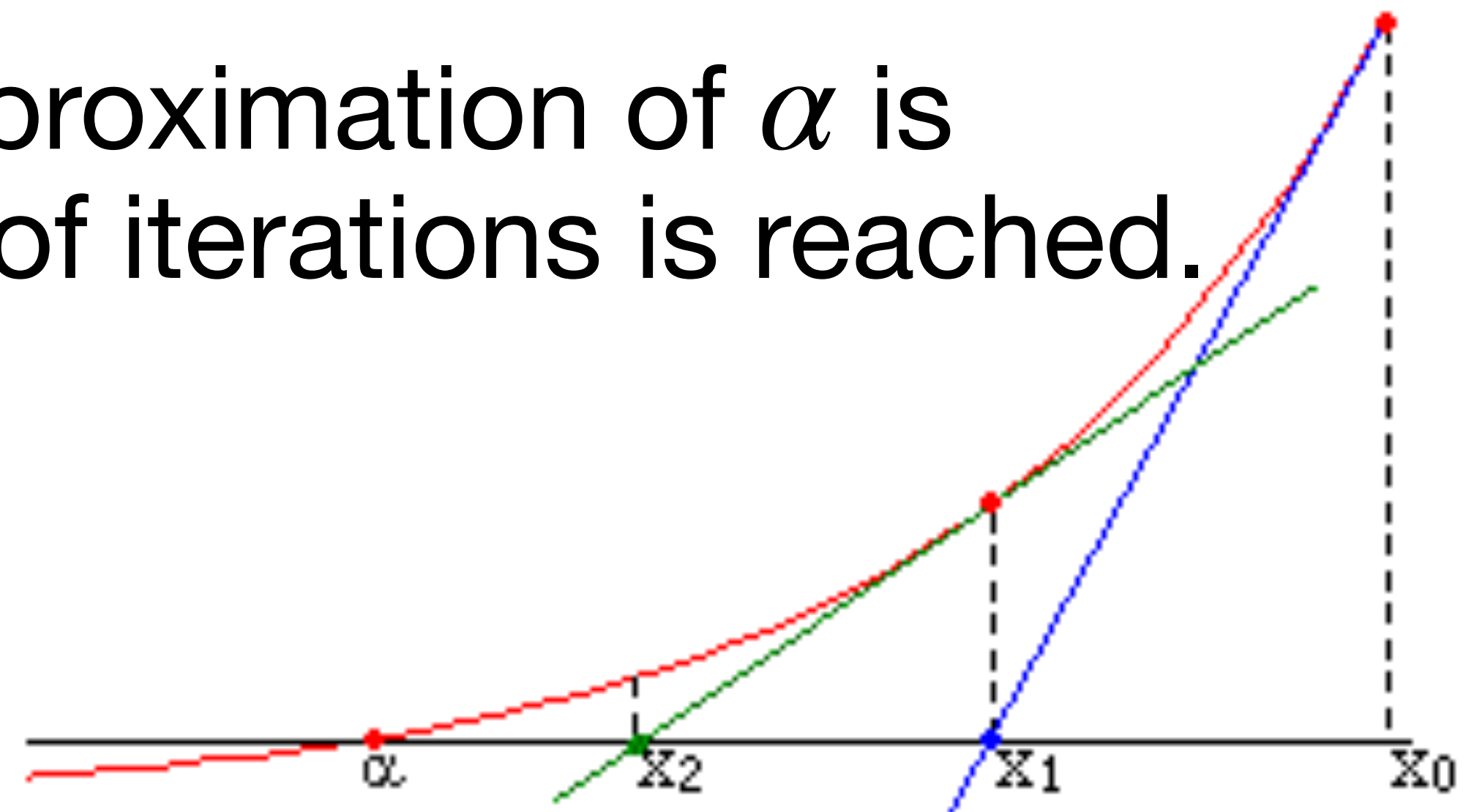
Notice: Geometrically, x_1 is exactly the root of the tangent line to f at the point x_0 .

Newton's method -2

Repeat this process as follows:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Stopping when an sufficiently good approximation of α is obtained or when a maximum number of iterations is reached.



Contrasting Newton and Bisection

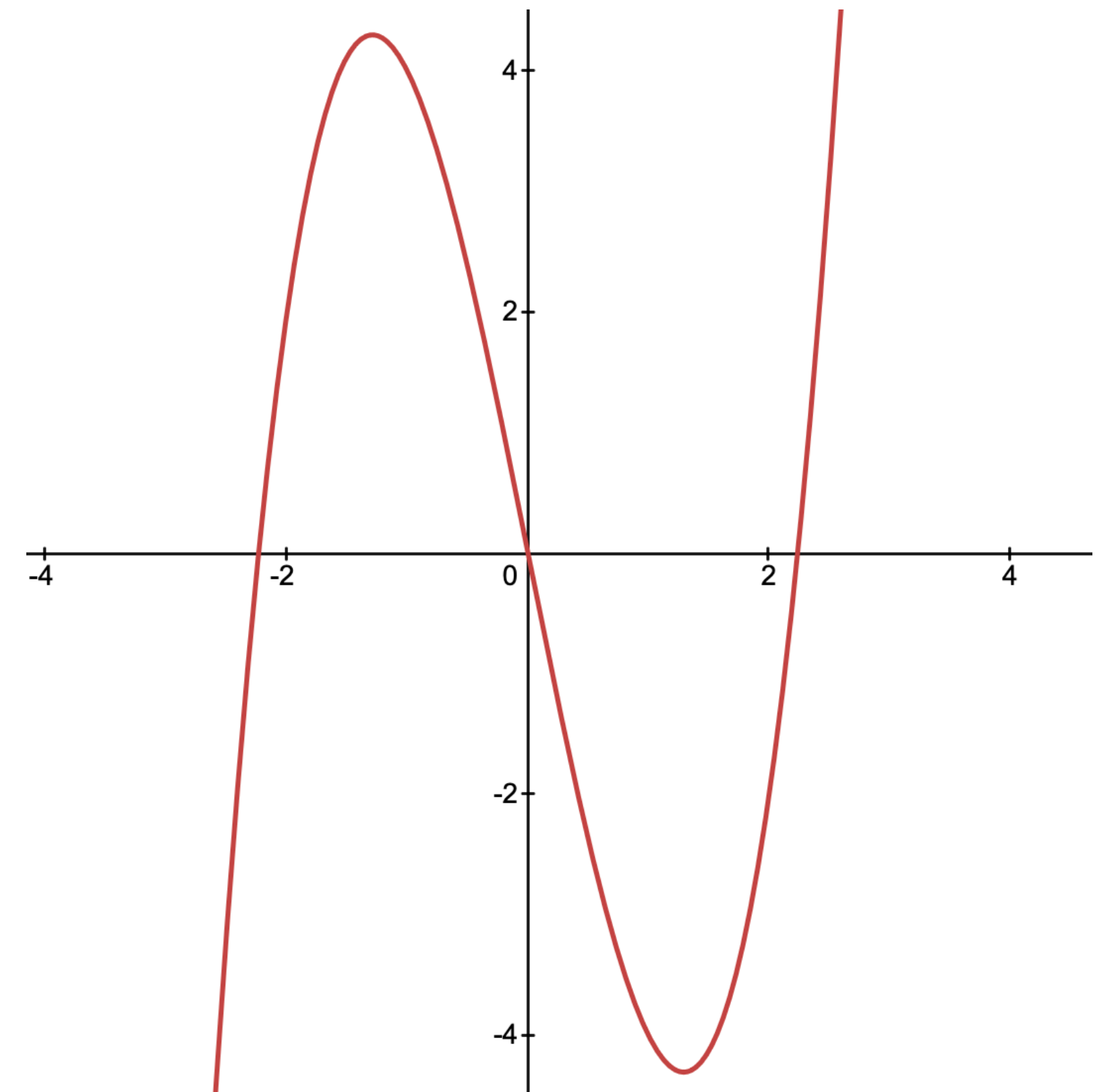
Newton's method comes with an extra requirement that bisection does not ask for. Namely, the existence of a way to compute derivatives. As this is a relatively big ask, Newton's method should give us some benefits for this. Some benefits of Newton's method are:

- Faster convergence. Bisection converges in linear time, that is each step reduces the length of the interval (and as such the error) by a factor of one half. Newton's method "can" converge quadratically.
- Newton's method can find roots of even multiplicity.

Contrasting Newton and Bisection

Newton's method does however have some key drawbacks. Namely, computing derivatives accurately can be computationally difficult and Newton's method can fail to converge to a root.

Example: $f(x) = x^3 - 5x$ with
initial guess $x_0 = 1$.



Newton's method failure

Given $f(x) = x^3 - 5x$ with initial guess $x_0 = 1$. We can see that

$$x_1 = x_0 - \frac{(1)^3 - 5(1)}{3(1)^2 - 5} = 1 - 2 = -1$$

$$x_2 = x_1 - \frac{(-1)^3 - 5(-1)}{3(-1)^2 - 5} = -1 + 2 = 1$$

This will oscillate endlessly.

Now go code up Newton's Method

A variant of Newton's method

A main drawback of Newton's method is that we need to know how to compute the derivative of f .

Later in this course, we will discuss at length how to compute the derivatives of functions (at least numerically). However, there does however exist a variant of Newton's method that does not require us to know how to compute the derivative of f .

Secant method - 1

Secant method works identically to Newton's method, except we replace the derivative of f at a point, with the slope of a closely related secant line about this point. That is, here

$$x_{n+1} = \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Note: In order to approximate a root with the secant method, we need to provide two initial guesses (in order to compute the secant line).

Secant method - 2

The formula in the previous slide for the n -th approximation of the secant method comes from the formula for the secant line. Recall that the equation of the secant line to f at points $a \leq b$ is given by:

$$y = \frac{f(b) - f(a)}{b - a}(x - b) + f(b)$$

Thus, use the secant line as an approximation of the tangent line to f at a point $c \in [a, b]$ (this is a reasonable approximation if $[a, b]$ is small!)